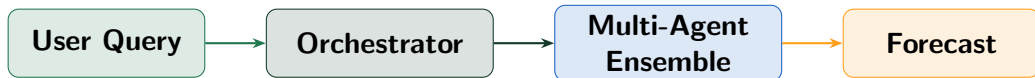


MandiSense AI

Intelligent Agricultural Price Forecasting



"Tomato in Kolar Mandi" — End-to-End Trace

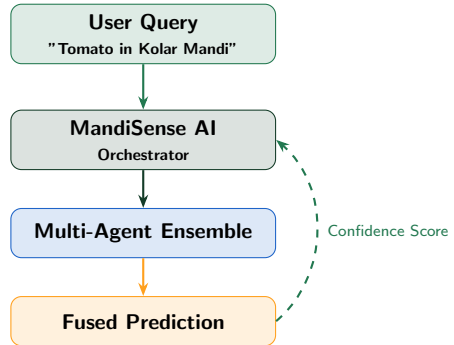
MandiSense AI — Intelligent Agricultural Forecasting

Mission

- **Multi-Agent Ensemble** for Mandi Price Prediction
- Highly accurate, **interpretable** price forecasts
- Hybrid architecture: seasonality + supply shocks + external signals
- Case Study: *"Tomato price for Kolar Mandi"*

Code Entry Point

- `run_agents.py` → `main()`



Problem & Objective

The Problem

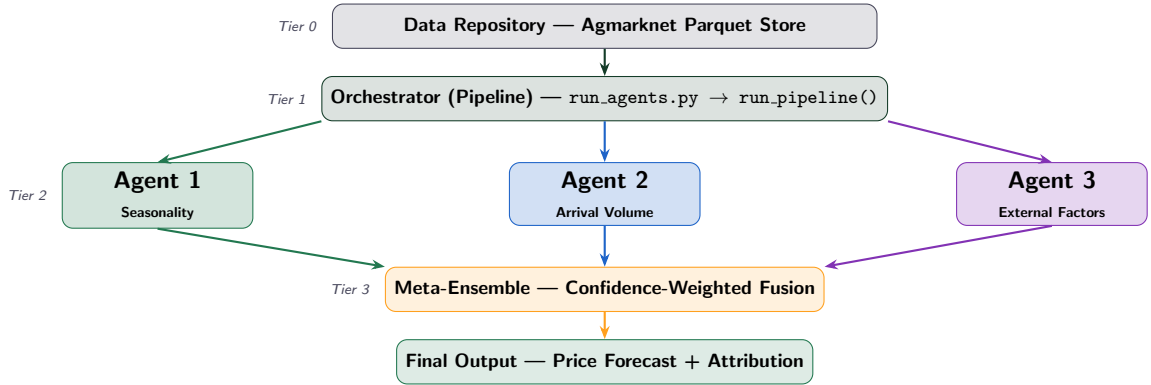
- Agricultural markets are **volatile**
- Disjoint factors: weather, festivals, supply
- Standard ML treats everything as one time series
- Cannot answer: *"Why is Tomato expensive today?"*



The Objective

- System that **reasons**, not just predicts
- 7-day and 30-day **explainable** forecasts
- Decompose complexity into specialist

System Architecture Overview



Code Mapping:

- `mandisense_ai/orchestrator/`
- `run_agents.py` → `run_pipeline()`

Key Properties:

- Agents execute **in parallel**
- Each agent is **independently replaceable**

External Factors Agent — The News & Weather Eye

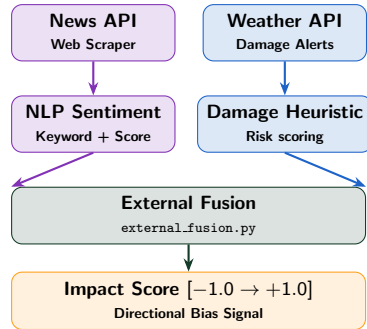
Agent Goal

Capture **non-numeric** market impacts

Data Sources

- **News scraper:** "Tomato Price Hike", "Rain damage"
- **Weather API:** Flood/drought alerts in Kolar region
- NLP Sentiment extraction
- Output: **Impact Score**
 $\in [-1.0, +1.0]$

Code Mapping:



Agent Outputs — Comparison & Conflict

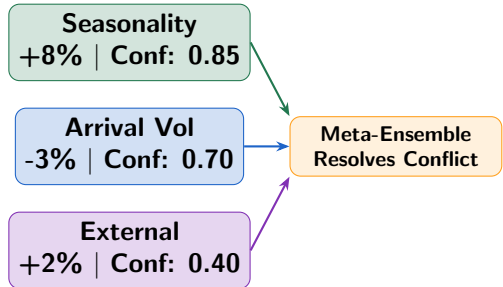
Agent	Pred.	Conf.	Signal
Seasonality	+8%	0.85	↑ Bullish
Arrival Vol	-3%	0.70	↓ Bearish
External	+2%	0.40	↑ Positive

[!] Conflict Detected

Seasonality (+8%) **vs.** Arrival (-3%)
High-confidence agents disagree on direction!

Code Mapping:

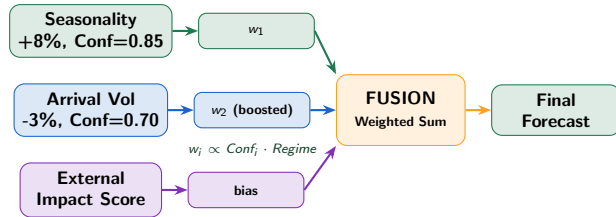
- `run_agents.py` Lines 31–47
- Captures `seasonality_output`, `arrival_output`, `external_impact`



Meta-Ensemble — Confidence-Aware Fusion

Fusion Logic

- **Dynamic Weighting:**
High-confidence agent gets more weight
- **Arrival Boost:** If supply stress is critical, w_2 is amplified
- **Horizon Alignment:** Normalises 30-day seasonal to 7-day scale
- External acts as a **directional bias**, not a forecast



Code Mapping:

- `meta_ensemble.py` → `fuse()`
(Line 192)

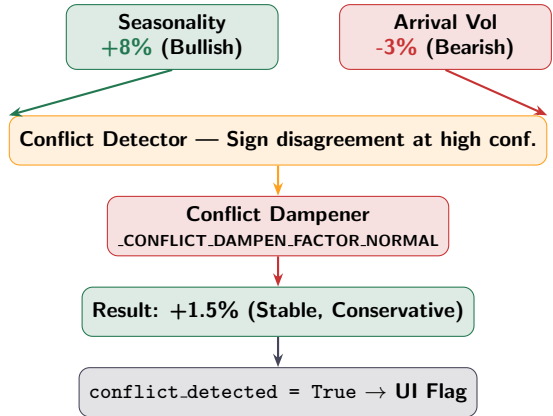
Conflict Handling — The Safety Net

Conflict Detection Logic

- Two **high-confidence** agents point in **opposite directions**
- System **dampens** the result
- Prevents extreme, risky predictions
- Sets `conflict_detected=True` for UI

Design Principle

"We'd rather be slightly wrong than confidently wrong."



Agent Outputs — Comparison & Conflict

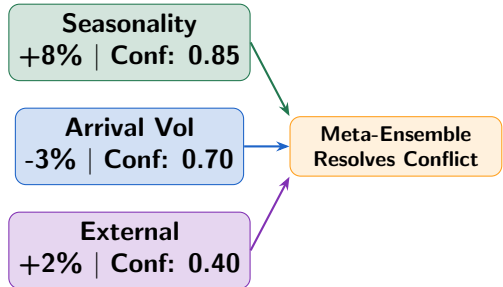
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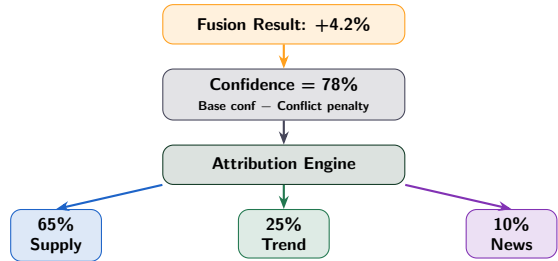
- `run_agents.py` Lines 31–47
- Captures `seasonality_output`, `arrival_output`, `external_impact`



Final Prediction Generation

Prediction Output

- **Numerical Fusion:** Final % change calculated
- **Confidence:** Aggregated with conflict penalty applied
- **Attribution:** Explains *which* agent drove the result



[~] Sample Output — Tomato / Kolar

- Prediction: **+4.2%**
- Confidence: **78%**
- Supply (Arrival): 65%
- Trend (Seasonal): 25%

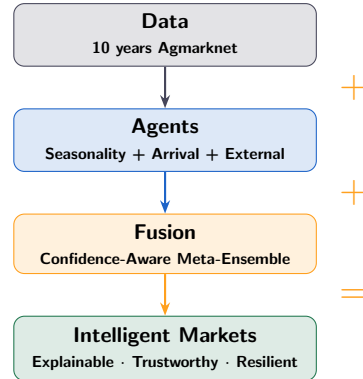
Conclusion & Future Outlook

Current State

- Stable **3-agent ensemble** for 5+ commodities
- Full end-to-end pipeline from query to decision
- Conflict-aware, confidence-weighted fusion
- Explainable attribution at every level

Future Roadmap

- **Causal Inference** — "Why exactly?"
- **LLM Reasoning Traces** — Natural language explanations



"MandiSense AI is the intersection of Data Science and Domain Expertise."

Thank You

MandiSense AI — Phase 1 Review

BMS College of Engineering